Classification of Pipe Conditions

CE 4318- Machine Learning for CE

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"A leak-proof XGBoost pipeline with custom robustness diagnostics"

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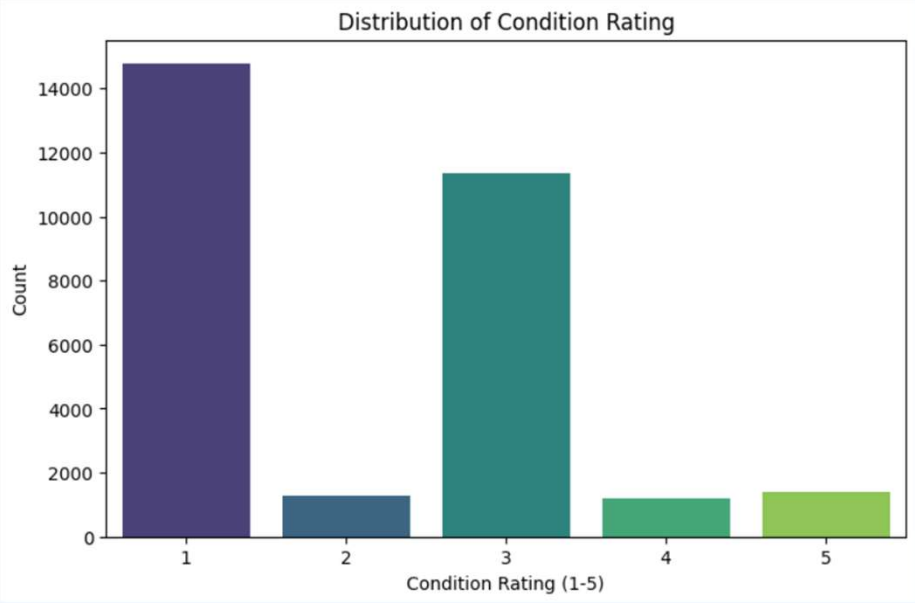
01 Context/Objectives

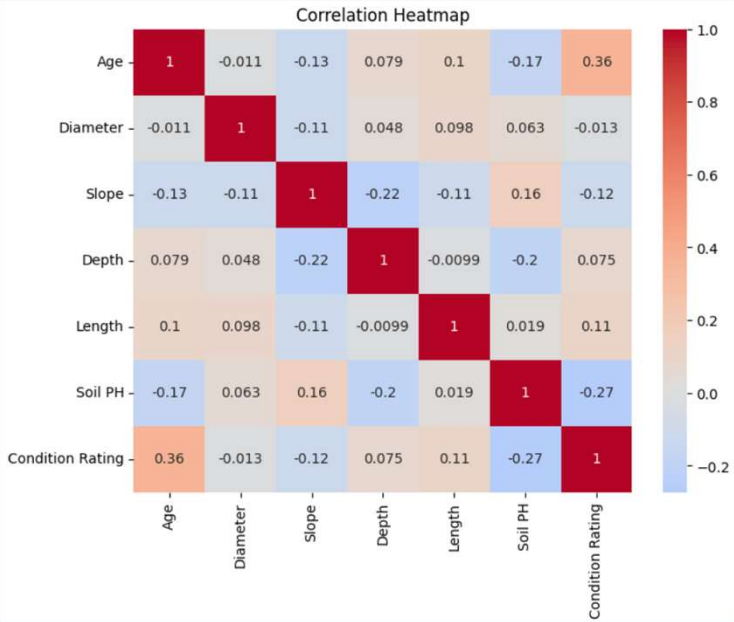
- **Why this matters:** Municipal water networks are aging; reactive maintenance is 5-10x more expensive than proactive replacement.
- **Objective:** Predict Condition Rating (1–5) using physical and environmental features to prioritize inspections.
- **Constraints:** Synthetic municipal dataset, mixed feature types, and mild class imbalance.

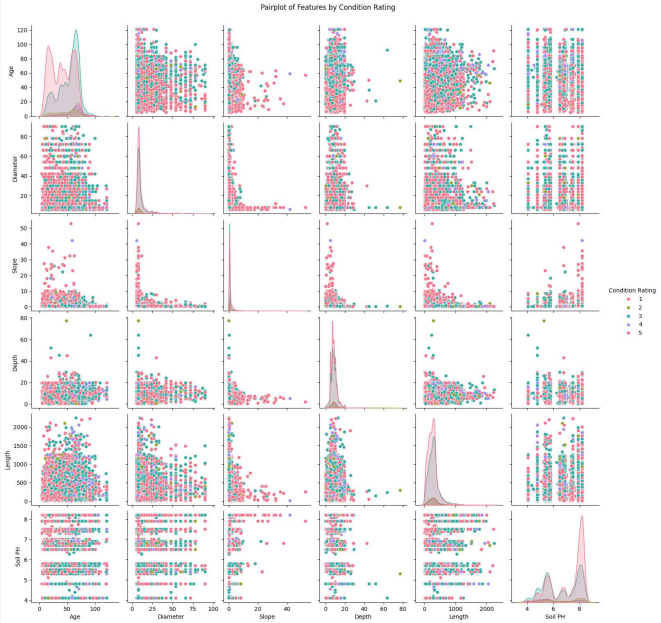
02 Data Exploration

- **Numerical:** Age, Soil PH
- **Skewed Numerical:** Diameter, Slope, Depth, Length #Log Transform
- **Categorical:** Material, Soil Type, Road Type
- **Engineered Interactions:** Age × Soil PH, Age x Material, Length x Slope
- **Target:** Condition Rating (1–5), Ordinal



Heat Map & Pairplot

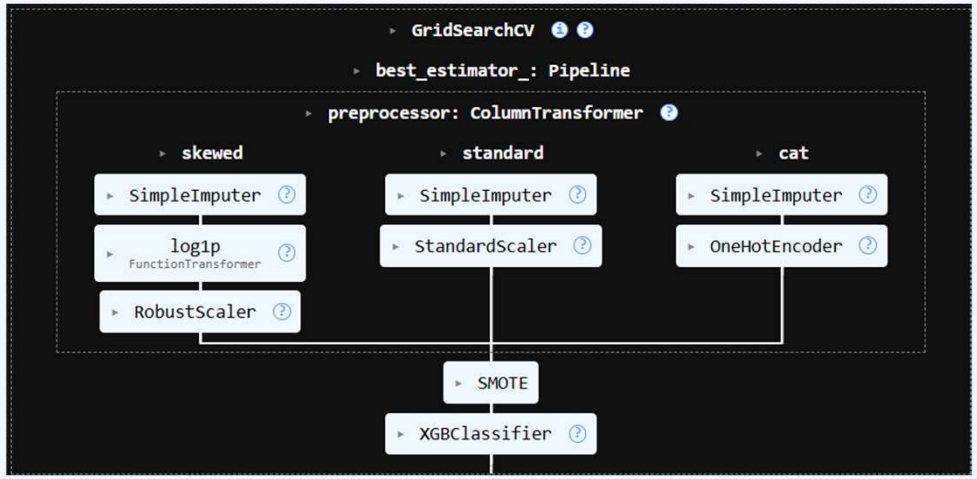




- No Major Correlation Issues
- Lots of the physical dimension-based features are heavily skewed

03 Pipeline Architecture

- **Preprocessing Branches:**
- Skewed Numerics → Median Imputation + StandardScaler
- Standard Numerics → Mean Imputation + StandardScaler
- Categoricals → Most-Frequent Imputation + OneHotEncoder
- **The Leak-Proof Flow:** > ColumnTransformer → SMOTE (Training Data Only) → XGBoost Classifier



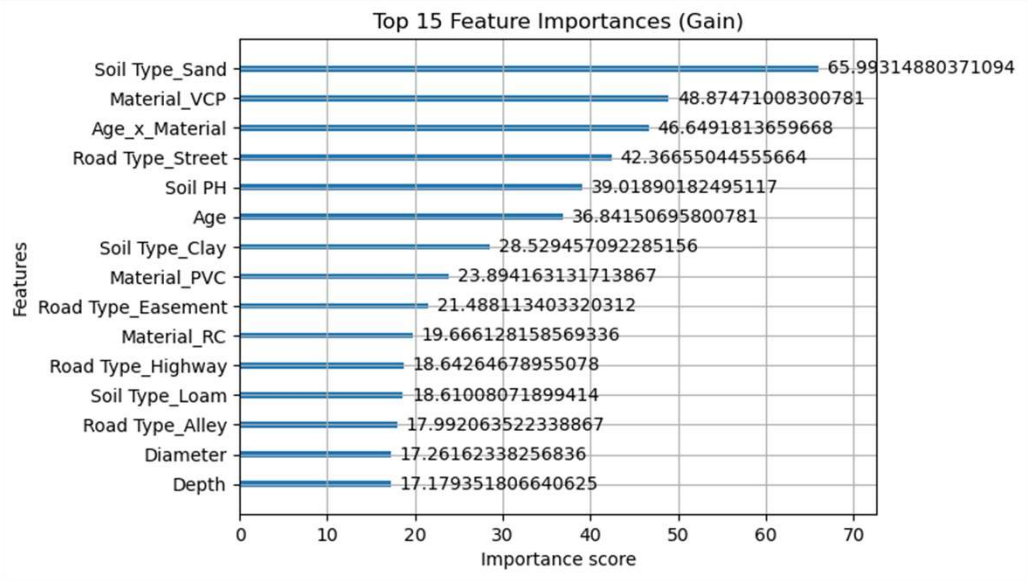
#SMOTE (Synthetic Minority Over-sampling): Balances the dataset by creating *new, synthetic* examples of rare pipe conditions (e.g., ratings 1 or 5) rather than just duplicating old data

04 Model Training

- **Model:** XGBoost Classifier
- **Objective:** multi:softmax (5 Discrete Classes)
- **Evaluation Metric:** mlogloss
- **Key Parameters (GridSerchCV):** Stratified

Criteria = F1 macro
'model__colsample_bytree': 0.8,
'model__learning_rate': 0.1,
'model__max_depth': 3,
'model__n_estimators': 300,
'model__subsample': 0.8}

Joblib to save model

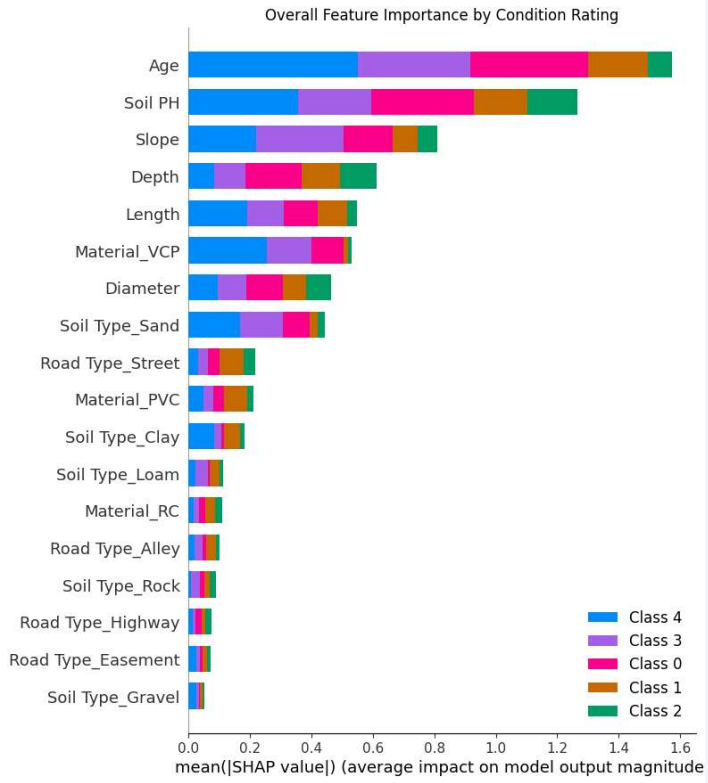


Top Features: Sand, Soil PH, and VCP

SHapley Additive exPlanations

- **Age/Soil are top universal condition drivers.**
- **Environment beats material type.**

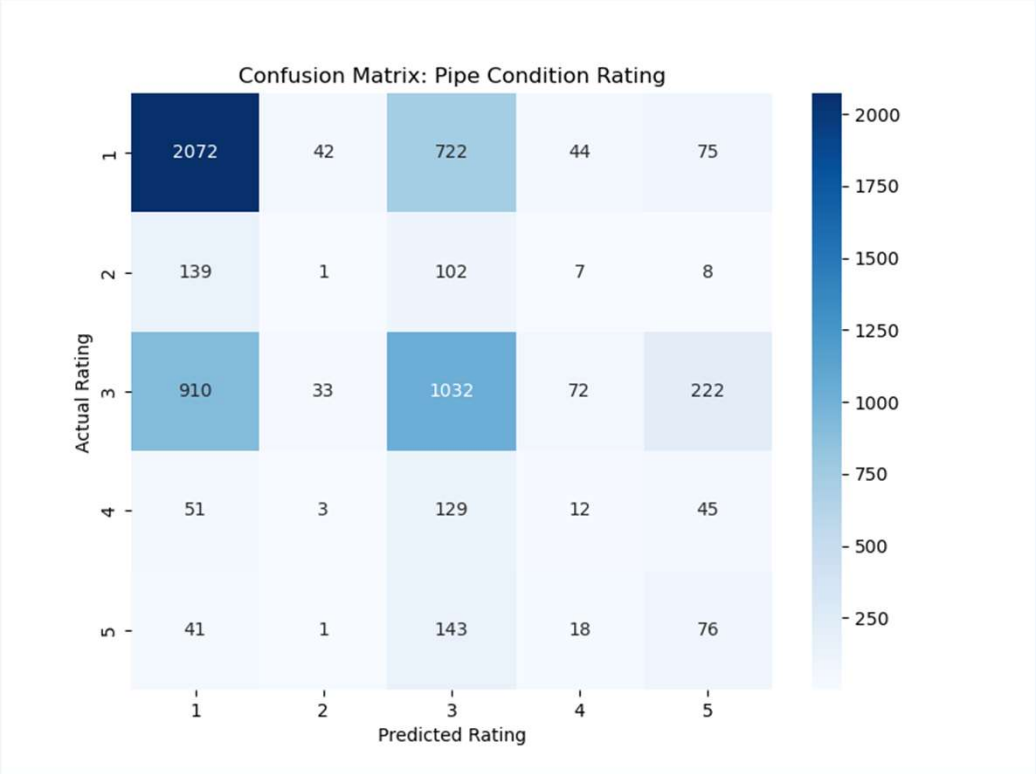
#SHAP (Model Transparency): SHAP acts as a "decoder" that quantifies exactly how much each individual feature like a pipe's age or soil acidity contributed to the model's final condition rating.



05 Evaluation

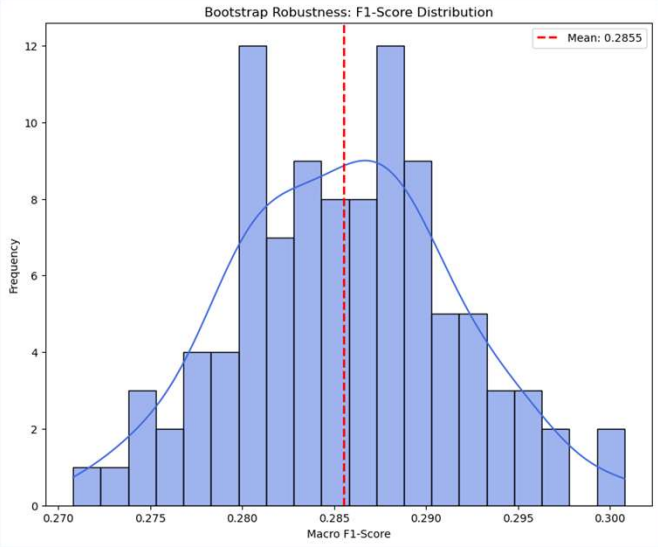
- **Accuracy:** 0.5322
- **Mean Absolute Error (MAE):** 0.8950
- **Quadratic Weighted Kappa (QWK):** 0.3797

The model heavily biases predictions toward classes 1 and 3, struggling to identify other classes.



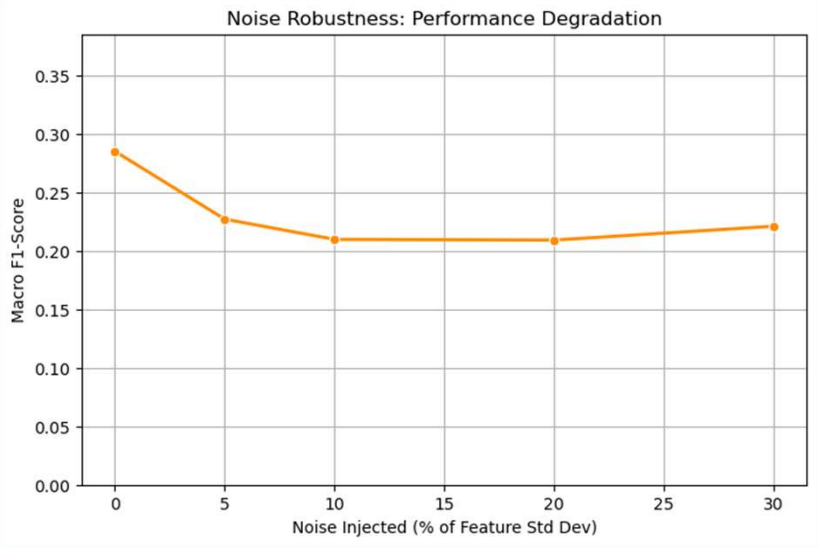
05 Robustness Check

Bootstrap:



Stable- tight distribution around the mean of **0.2785**
CL=95% , 100 resamples of test set

Noise:



Highly resilient to noisy data

06 Conclusion

Key Takeaways:

•Environmental factors are the primary drivers of pipe degradation, with Age and Soil PH impacting conditions more heavily than the pipe's physical material.

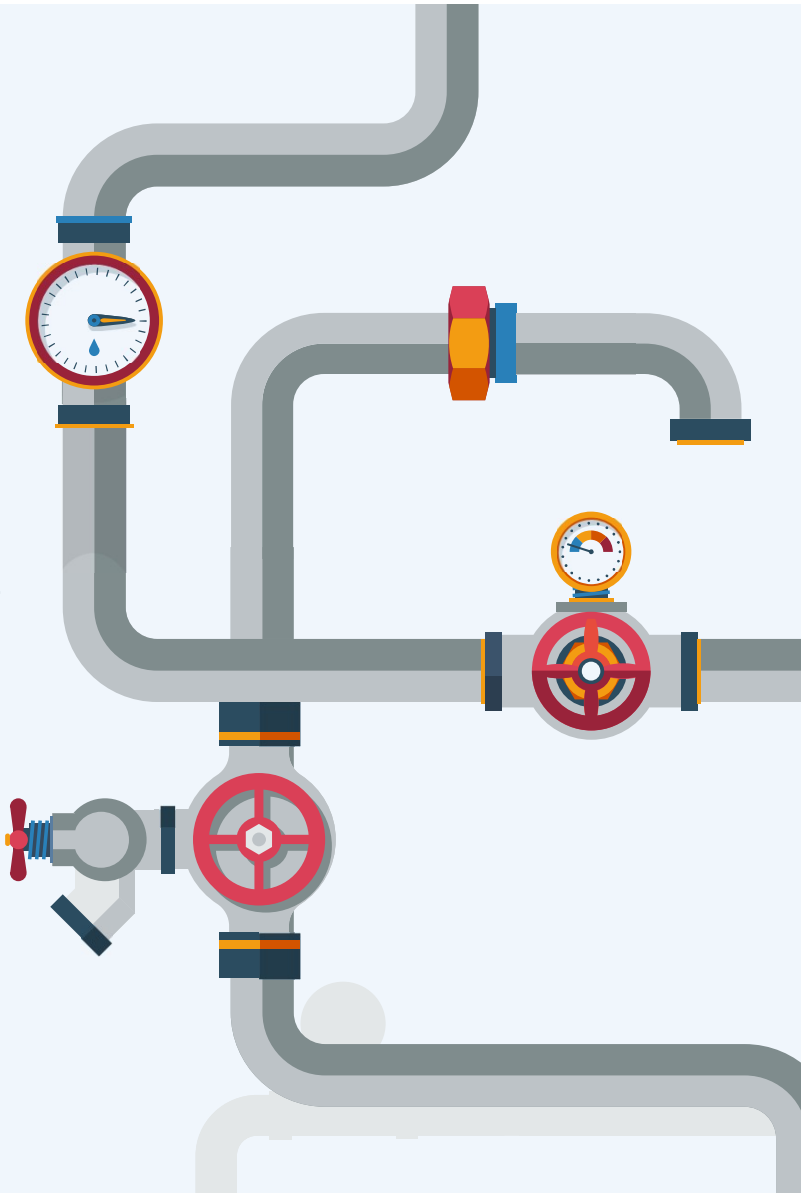
•Not enough distinct variance to separate borderline conditions even with SMOTE.

Area for Improvement:

•Despite utilizing SMOTE, the model still struggles with class imbalance and heavily biases its predictions toward classes 1 and 3. Future iterations could explore adjusting class weights or utilizing different sampling techniques to improve the model's 53.22% accuracy and better capture minority classes.

•Could group conditions into “Urgent Maintenance (1-2)” and “Postpone (3-5)”.

Thanks!

Repo:
https://github.com/mazzenawad/CE4318_Pipe Conditions_ML_Pipeline_Project

Do you have any questions?